

Unit 4 - Orthogonalization, Eigen Values & Eigen Vectors

Lectures 18/2

Topic 1 - Orthogonalization - Orthogonal matrices, Properties, Rectangular matrices with orthonormal columns

→ In an orthogonal basis, every vector is perpendicular to every other vector. The coordinate axes are mutually orthogonal.

→ Mutually perpendicular unit vectors are called Orthonormal vectors.

→ For the vector space $\mathbb{R}^2$,

1. The set $(2,0), (0,2)$ is an orthogonal basis.

2. The set $(1,-2), (2,1)$ is an orthogonal basis.

3. The set $(1,0), (0,1)$ is an orthonormal basis.

→ A matrix with orthonormal columns will be called Q .

→ A square matrix with orthonormal columns is called an Orthogonal matrix denoted by Q .

→ Example: Rotation matrix, Any permutation matrix.

Properties of Q :-

→ If Q (square or rectangular) has orthonormal columns, then $Q^T \cdot Q = I$.

②

→ An orthogonal matrix is a square matrix with orthonormal columns. Then $Q^T = Q^{-1}$

→ If Q is rectangular, then Q^T is left inverse of Q .

→ Multiplication by any Q preserves length. The norms of x & Qx are equal.

→ Also Q preserves inner products and angles,

$$\text{Since } (Qx)^T (Qy) = x^T Q^T Q y = x^T y$$

→ If $q_1, q_2 \dots q_n$ are orthonormal basis of $\mathbb{R}^n$ then any vector b from $\mathbb{R}^n$ can be expressed as

$$b = x_1 q_1 + x_2 q_2 + \dots + x_n q_n \quad \text{--- (1)}$$

Multiplying both sides by q_1^T

$$q_1^T \cdot b = x_1 q_1^T q_1 + x_2 \underbrace{q_1^T q_2}_0 + \dots + x_n \underbrace{q_1^T q_n}_0$$

$$\therefore q_1^T \cdot b = x_1$$

$$\text{Similarly } x_2 = q_2^T \cdot b \dots x_n = q_n^T \cdot b$$

$$\text{Hence } b = (q_1^T \cdot b) q_1 + (q_2^T \cdot b) q_2 + \dots + (q_n^T \cdot b) q_n$$

= Sum of one dimensional projections onto q_i 's.

→ The matrix form of (1) is $Qx = b$ and the solution of this system of equations is

$$x = Q^{-1} \cdot b$$

$$\boxed{x = Q^T \cdot b}$$

→ The rows of a square matrix are orthonormal whenever the columns are.

Orthonormal columns
Orthonormal rows

$$Q = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 0 & -2/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \end{bmatrix}$$

Rectangular matrices with orthonormal columns

→ If Q has orthonormal columns, the least squares problem becomes easy.

→ $Q^T Q x = Q^T b$ are the normal equations for the best solution in which $Q^T Q = I$.

→ $x = Q^T b$

→ $p = Qx$, the projection of b is

$$(a_1^T \cdot b) a_1 + \dots + (a_n^T \cdot b) a_n$$

→ $p = Q \cdot Q^T b$, the projection matrix is $P = Q \cdot Q^T$

Special cases

1) If Q is a square matrix, $\hat{x} = Q^T b$
 $\hat{x} = Q^{-1} \cdot b \quad \begin{bmatrix} 0 & 0 & Q^T = Q^{-1} \end{bmatrix}$

2) If Q is a square matrix, $P = Q \hat{x}$
 $P = Q \cdot Q^{-1} \cdot b$
 $\boxed{P = b}$

Example Project $b = (0, 3, 0)$ onto each of the orthonormal vectors $a_1 = (2/3, 2/3, -1/3)$; $a_2 = (-1/3, 2/3, 2/3)$ & then find its projection its projection P onto the plane of a_1 & a_2 .

(4)

Solution The projection of b onto a_1 is given by

$$P_1 = \left(\frac{a_1^T \cdot b}{a_1^T a_1} \right) a_1 \quad \text{but } a_1^T \cdot a_1 = \|a_1\|^2 = \sqrt{\frac{4}{9} + \frac{4}{9} + \frac{1}{9}} = 1$$

$$P_1 = \begin{bmatrix} 2/3 & 2/3 & -1/3 \end{bmatrix} \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix} * \begin{bmatrix} 2/3 \\ 2/3 \\ -1/3 \end{bmatrix} = 2 \begin{bmatrix} 2/3 \\ 2/3 \\ -1/3 \end{bmatrix}$$

$$P_1 = \begin{bmatrix} 4/3 \\ 4/3 \\ -2/3 \end{bmatrix}$$

Similarly, $P_2 = \frac{(a_2^T b) a_2}{a_2^T a_2}$; $a_2^T a_2 = \|a_2\|^2 = \sqrt{1/9 + 4/9 + 4/9} = 1$

$$P_2 = \begin{bmatrix} -1/3 & 2/3 & 2/3 \end{bmatrix} \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix} \begin{bmatrix} -1/3 \\ 2/3 \\ 2/3 \end{bmatrix} = \begin{bmatrix} -2/3 \\ 4/3 \\ 4/3 \end{bmatrix}$$

∴ P is the projection onto the plane containing a_1 & a_2 i.e. $P = P_1 + P_2 =$

$$\begin{bmatrix} 4/3 \\ 4/3 \\ -2/3 \end{bmatrix} + \begin{bmatrix} -2/3 \\ 4/3 \\ 4/3 \end{bmatrix} = \begin{bmatrix} 2/3 \\ 8/3 \\ 2/3 \end{bmatrix}$$

Example Find a third column so that the matrix is orthogonal. Verify that the rows automatically become orthonormal at the same time.

$$Q = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{14} & - \\ 1/\sqrt{3} & 2/\sqrt{14} & - \\ 1/\sqrt{3} & -3/\sqrt{14} & - \end{bmatrix}$$

Solution By the definition of orthogonal matrix, first and second columns are orthogonal to the required third column. Let the third column be $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$. (5)

$$Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{14}} & x \\ \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{14}} & y \\ \frac{1}{\sqrt{3}} & \frac{-3}{\sqrt{14}} & z \end{bmatrix} \Rightarrow a^T c = 0$$

$\underbrace{\quad}_a \quad \underbrace{\quad}_b \quad \underbrace{\quad}_c$

$$\begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$\frac{x}{\sqrt{3}} + \frac{y}{\sqrt{3}} + \frac{z}{\sqrt{3}} = 0$$

$$\Rightarrow x + y + z = 0 \quad \text{--- (1)}$$

Similarly, $b^T c = 0 \Rightarrow \begin{bmatrix} \frac{1}{\sqrt{14}} \\ \frac{2}{\sqrt{14}} \\ \frac{-3}{\sqrt{14}} \end{bmatrix} \Rightarrow \begin{bmatrix} \frac{1}{\sqrt{14}} & \frac{2}{\sqrt{14}} & \frac{-3}{\sqrt{14}} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$

$\underbrace{\quad}_b$

$$\frac{x}{\sqrt{14}} + \frac{2y}{\sqrt{14}} - \frac{3z}{\sqrt{14}} = 0 \Rightarrow x + 2y - 3z = 0$$

$$x + y + z = 0$$

$$x + 2y - 3z = 0$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \end{bmatrix} \xrightarrow{R_2 - R_1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -4 \end{bmatrix}$$

$$x + y + z = 0$$

$$y - 4z = 0$$

$$y = 4z$$

$$x + 4z + z = 0$$

$$x = -5z$$

$$y = 4z$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -5z \\ 4z \\ z \end{pmatrix} = z \begin{pmatrix} -5 \\ 4 \\ 1 \end{pmatrix}$$

⑥

→ For rows to be orthonormal, the normal must be equal to 1.

$$\left(\frac{1}{\sqrt{3}}\right)^2 + \left(\frac{1}{\sqrt{14}}\right)^2 + (-5z)^2 = 1$$

$$\frac{1}{3} + \frac{1}{14} + 25z^2 = 1$$

$$25z^2 = \frac{1}{1} - \frac{1}{3} - \frac{1}{14}$$

$$25z^2 = \frac{42 - 14 - 3}{42}$$

$$25z^2 = \frac{25}{42}$$

$$z^2 = 1/42 \Rightarrow z = \frac{1}{\sqrt{42}}$$

$${}^o_o \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{bmatrix} -5/\sqrt{42} \\ 4/\sqrt{42} \\ 1/\sqrt{42} \end{bmatrix}$$

$${}^o_o Q = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{14} & -5/\sqrt{42} \\ 1/\sqrt{3} & 2/\sqrt{14} & 4/\sqrt{42} \\ 1/\sqrt{3} & -3/\sqrt{14} & 1/\sqrt{42} \end{bmatrix}$$

A = QR factorizationThe gram-schmidt process

→ It is a process of converting linearly independent vectors into orthonormal vectors.

→ consider any 3 independent vectors a, b, c . Then the first orthonormal vector $a_1 = \frac{a}{\text{norm}\{a\}}$

→ If 'b' is perpendicular to the vector 'a' then

$$a_2 = \frac{b}{\text{norm}(b)} \text{ otherwise } B = b - (a_1^T b) a_1 \text{ and}$$

$$a_2 = \frac{B}{\text{norm}(B)}$$

→ If 'c' is perpendicular to the plane spanned by the vectors 'a' and 'b' then $a_3 = \frac{c}{\text{norm}(c)}$ otherwise

$$C = c - (a_1^T c) a_1 - (a_2^T c) a_2 \text{ and } a_3 = \frac{C}{\text{norm}(C)}$$

→ This is the idea of the whole Gram-Schmidt process, to subtract from every new vector its components in the directions that are already settled. That idea is used over & over again. When there is a fourth vector, we subtract away its components in the directions of a_1, a_2, a_3 .

Example From the vectors a, b, c find orthonormal vectors a_1, a_2 and a_3 given $a = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ $b = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ $c = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$

Solution From Gram-Schmidt Process

$$a_1 = \frac{a}{\|a\|} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

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$$a_{r2} = \frac{e_2}{\|e_2\|}$$

$$e_2 = b - (a_{r1}^T b) a_{r1}$$

$$e_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{pmatrix}$$

$$e_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{\sqrt{2}} \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{pmatrix}$$

$$e_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 1/\sqrt{2} * \frac{1}{\sqrt{2}} \\ 1/2 \\ 0 \end{pmatrix}$$

$$e_2 = \begin{pmatrix} 1/2 \\ -1/2 \\ 1 \end{pmatrix}$$

$$a_{r2} = \frac{e_2}{\|e_2\|} = \frac{1}{\sqrt{1/4 + 1/4 + 1}} \cdot \begin{pmatrix} 1/2 \\ -1/2 \\ 1 \end{pmatrix}$$

$$= \frac{1}{\sqrt{\frac{1+1+4}{4}}} \begin{pmatrix} 1/2 \\ -1/2 \\ 1 \end{pmatrix}$$

$$a_{r2} = \frac{2}{\sqrt{6}} \begin{pmatrix} 1/2 \\ -1/2 \\ 1 \end{pmatrix} = \begin{bmatrix} 1/\sqrt{6} \\ -1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}$$

$$a_3 = \frac{e_3}{\|e_3\|} \quad e_3 = c - (a_1^T c) a_1 - (a_2^T c) a_2$$

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$$e_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{pmatrix}$$

$$- \begin{bmatrix} 1/\sqrt{6} & -1/\sqrt{6} & 2/\sqrt{6} \end{bmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \begin{bmatrix} 1/\sqrt{6} \\ -1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}$$

$$a_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \frac{1}{\sqrt{2}} \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{pmatrix} - \frac{1}{\sqrt{6}} \begin{bmatrix} 1/\sqrt{6} \\ -1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix} = e_3$$

$$a_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 1/2 \\ 1/2 \\ 0 \end{pmatrix} - \begin{pmatrix} 1/6 \\ -1/6 \\ 2/6 \end{pmatrix} = e_3$$

$$e_3 = \begin{pmatrix} -1/2 - 1/6 \\ 1/2 - 1/2 + 1/6 \\ 1/2 - 2/6 \end{pmatrix} = \begin{pmatrix} \frac{-3-1}{6} \\ \frac{6-3+1}{6} \\ \frac{6-2}{6} \end{pmatrix} = \begin{pmatrix} -2/3 \\ 2/3 \\ 2/3 \end{pmatrix}$$

$$a_3 = \frac{e_3}{\|e_3\|} = \frac{1}{\sqrt{4/9 + 4/9 + 4/9}} \begin{pmatrix} -2/3 \\ 2/3 \\ 2/3 \end{pmatrix} = \frac{\sqrt{3}}{\sqrt{4}} \begin{pmatrix} -2/3 \\ 2/3 \\ 2/3 \end{pmatrix}$$

$$\therefore a_3 = \begin{pmatrix} -1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix}$$

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A = QR factorization :-

→ Let A be a matrix whose columns are a, b, c

→ Let Q be the matrix whose columns are q_1, q_2 and q_3 which are determined using Gram Schmidt process.

→ Then to find the third matrix which connects A & Q, express a, b, c as a linear combination of q_1, q_2, q_3 .

→ The whole factorization is :-

$$A = [a \ b \ c] = [q_1 \ q_2 \ q_3] \begin{bmatrix} q_1^T a & q_1^T b & q_1^T c \\ & q_2^T b & q_2^T c \\ & & q_3^T c \end{bmatrix}$$

$$A = Q \cdot R$$

Note :- consider the normal equation for $Ax = b$

$$A^T \cdot A \cdot \hat{x} = A^T b \quad \text{But } A = QR$$

$$(QR)^T (QR) \hat{x} = (QR)^T b$$

$$\underbrace{R^T \cdot Q^T \cdot Q \cdot R}_{I} \cdot \hat{x} = R^T \cdot Q^T \cdot b$$

$$R^T \cdot R \cdot \hat{x} = R^T \cdot Q^T \cdot b$$

$$R \hat{x} = Q^T b$$

⇒ When $Ax = b$ is not solvable, we consider $R \hat{x} = Q^T b$ and solve.

→ If A is a square matrix of order 'n' then Q & R are also square matrix of order 'n'.

But if A is a matrix of order ' $m \times n$ ' then Q is also a matrix of order ' $m \times n$ ' but R is a square matrix of order ' n '.

$$\begin{bmatrix} \uparrow & \uparrow \\ a & b \\ \downarrow & \downarrow \end{bmatrix} = \begin{bmatrix} \uparrow & \uparrow \\ q_1 & q_2 \\ \downarrow & \downarrow \end{bmatrix} \begin{bmatrix} q_1^T a & q_1^T b \\ 0 & q_2^T b \end{bmatrix}$$

Example Factorize $\begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & 0 \end{bmatrix}$ into QR

recognizing that first column is already a unit vector.

Solution:- $A = \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & 0 \end{bmatrix}$ $Q = [q_1 \ q_2]$

$\downarrow$ $\downarrow$
 a b

$$q_1 = \frac{a}{\|a\|} = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$$

$$q_2 = \frac{e_2}{\|e_2\|}; \quad e_2 = b - (q_1^T b) q_1$$

$$e_2 = \begin{pmatrix} \sin \theta \\ 0 \end{pmatrix} - (\cos \theta \sin \theta) \begin{pmatrix} \sin \theta \\ 0 \end{pmatrix} \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$$

$$e_2 = \begin{pmatrix} \sin \theta \\ 0 \end{pmatrix} - (\cos \theta \cdot \sin \theta) \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$$

$$e_2 = \begin{pmatrix} \sin \theta \\ 0 \end{pmatrix} - \begin{pmatrix} \cos^2 \theta \cdot \sin \theta \\ \cos \theta \cdot \sin^2 \theta \end{pmatrix}$$

$$e_2 = \begin{pmatrix} \sin \theta (1 - \cos^2 \theta) \\ -\cos \theta \cdot \sin^2 \theta \end{pmatrix}$$

$$\|e_2\| = \sqrt{\{\sin \theta - \sin \theta \cdot \cos^2 \theta\}^2 + \cos^2 \theta \cdot \sin^4 \theta}$$

$$= \sqrt{\sin^2 \theta + \sin^2 \theta \cos^4 \theta - 2 \sin^2 \theta \cdot \cos^2 \theta + \cos^2 \theta \cdot \sin^4 \theta}$$

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$$\|e_2\| = \sqrt{\sin^2 \theta + \sin^2 \theta \cdot \cos^4 \theta - 2 \sin^2 \theta \cdot \cos^2 \theta + \cos^2 \theta \cdot \sin^4 \theta}$$

$$\|e_2\| = \sqrt{\sin^2 \theta + \sin^2 \theta \cdot \cos^2 \theta (\cos^2 \theta + \sin^2 \theta) - 2 \sin^2 \theta \cdot \cos^2 \theta}$$

$$\|e_2\| = \sqrt{\sin^2 \theta + \sin^2 \theta \cdot \cos^2 \theta - 2 \sin^2 \theta \cos^2 \theta}$$

$$\|e_2\| = \sqrt{\sin^2 \theta - \sin^2 \theta \cdot \cos^2 \theta}$$

$$\|e_2\| = \sqrt{\sin^2 \theta (1 - \cos^2 \theta)}$$

$$\|e_2\| = \sqrt{\sin^2 \theta \cdot \sin^2 \theta}$$

$$\|e_2\| = \sin^2 \theta$$

$$\therefore a_2 = \frac{e_2}{\|e_2\|} = \frac{1}{\sin^2 \theta} \begin{bmatrix} \sin \theta (1 - \cos^2 \theta) \\ -\cos \theta \cdot \sin^2 \theta \end{bmatrix}$$

$$\therefore a_2 = \begin{bmatrix} \sin \theta \\ -\cos \theta \end{bmatrix}$$

$$R = \begin{bmatrix} a_1^T a & a_1^T b \\ 0 & a_2^T b \end{bmatrix}; a_1 = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} a_2 = \begin{bmatrix} \sin \theta \\ -\cos \theta \end{bmatrix}$$

$$a_1^T a = [\cos \theta \sin \theta] \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} = 1$$

$$a_1^T b = [\cos \theta \sin \theta] \begin{bmatrix} \sin \theta \\ 0 \end{bmatrix} = \cos \theta \cdot \sin \theta$$

$$a_2^T b = [\sin \theta -\cos \theta] \begin{bmatrix} \sin \theta \\ 0 \end{bmatrix} = \sin^2 \theta$$

$$\therefore R = \begin{bmatrix} 1 & \cos \theta \cdot \sin \theta \\ 0 & \sin^2 \theta \end{bmatrix}$$

$$\therefore A = \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & 0 \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix} \begin{bmatrix} 1 & \cos \theta \cdot \sin \theta \\ 0 & \sin^2 \theta \end{bmatrix}$$

$$A = \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & 0 \end{bmatrix} = \begin{bmatrix} \cos \theta (\cos^2 \theta \cdot \sin \theta + \sin^3 \theta) \\ \sin \theta (\cos \theta \sin^2 \theta - \cos \theta \sin^2 \theta) \end{bmatrix} \quad (13)$$

$$A = \begin{bmatrix} \cos \theta & \sin \theta (\cos^2 \theta + \sin^2 \theta) \\ \sin \theta & 0 \end{bmatrix}$$

$$\therefore A = QR = \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & 0 \end{bmatrix}$$

Example Find an orthonormal set $\alpha_1, \alpha_2, \alpha_3$ for which α_1, α_2 span the column space of $A = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix}$.

What fundamental subspace contains α_3 . What is the least squares solution of $Ax = b$ if $b = [1 \ 2 \ 7]^T$?

solution $a = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$ $b = \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix} \Rightarrow \alpha_1 = \frac{a}{\|a\|} = \frac{1}{3} \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$

$$\therefore \alpha_1 = \begin{pmatrix} 1/3 \\ 2/3 \\ -2/3 \end{pmatrix}$$

$$\alpha_2 = \frac{e_2}{\|e_2\|} ; e_2 = b - (\alpha_1^T b) \alpha_1$$

$$e_2 = \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix} - \begin{pmatrix} 1/3 & 2/3 & -2/3 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix} \begin{pmatrix} 1/3 \\ 2/3 \\ -2/3 \end{pmatrix}$$

$$e_2 = \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix} - \left(\frac{1}{3} - \frac{2}{3} - \frac{8}{3} \right) \begin{pmatrix} 1/3 \\ 2/3 \\ -2/3 \end{pmatrix}$$

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$$e_2 = \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix} - (-3) \begin{pmatrix} 1/3 \\ 2/3 \\ -2/3 \end{pmatrix}$$

$$e_2 = \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix} - \begin{pmatrix} -1 \\ -2 \\ +2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$$

$$a_2 = \frac{e_2}{\|e_2\|} = \frac{1}{\sqrt{9}} \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$$

$$\therefore a_2 = \begin{pmatrix} 2/3 \\ 1/3 \\ 2/3 \end{pmatrix}$$

Let $e_3 = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$. To find a_3 , let the vector be assumed

as orthogonal to the plane spanned by a_1 and a_2 such that $a_1^T \cdot e_3 = 0$ & $a_2^T \cdot e_3 = 0$

$$\begin{bmatrix} 1/3 & 2/3 & -2/3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0 \quad ; \quad \begin{bmatrix} 2/3 & 1/3 & 2/3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$x + 2y - 2z = 0$$

$$2x + y + 2z = 0$$

$$\begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \end{bmatrix} \xrightarrow{R_2 - 2R_1} \begin{bmatrix} 1 & 2 & -2 \\ 0 & -3 & 6 \end{bmatrix}$$

$$-3y + 6z = 0$$

$$x + 2y - 2z = 0$$

$$-3y = -6z$$

$$\boxed{y = 2z}$$

$$x + 2(2z) - 2z = 0$$

$$x + 4z - 2z = 0$$

$$x + 2z = 0$$

$$\boxed{x = -2z}$$

$$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -2z \\ 2z \\ z \end{pmatrix} = z \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix}$$

$$Q = \begin{pmatrix} 1/3 & 2/3 & -2z \\ 2/3 & 1/3 & 2z \\ -2/3 & 2/3 & z \end{pmatrix}$$

rows should be orthonormal

$$\circ \circ \quad \sqrt{1/9 + 4/9 + 4z^2} = 1$$

$$1/9 + 4/9 + 4z^2 = 1$$

$$4z^2 = 1 - \frac{5}{9}$$

$$4z^2 = 4/9$$

$$z^2 = 1/9$$

$$\boxed{z = 1/3}$$

$$\circ \circ \quad Q = \begin{bmatrix} 1/3 & 2/3 & -2/3 \\ 2/3 & 1/3 & 2/3 \\ -2/3 & 2/3 & 1/3 \end{bmatrix}$$

$$\text{consider } Ax = b \Rightarrow A = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}$$

We can verify that the above equation is not solvable & hence we consider $R\hat{x} = Q^T b$ where

$$R = \begin{bmatrix} a_1^T a & a_1^T b \\ 0 & a_2^T b \end{bmatrix}$$

$$a_1^T a = \begin{bmatrix} 1/3 & 2/3 & -2/3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} = 1/3 + 4/3 + 4/3 = 3$$

$$a_1^T b = \begin{bmatrix} 1/3 & 2/3 & -2/3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} = 1/3 - 2/3 - 8/3 = -3$$

$$(16) \quad a_2^T b = \begin{bmatrix} 2/3 & 1/3 & 2/3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} = 2/3 - \frac{1}{3} + \frac{8}{3} = 3$$

$$\circ \circ \quad R = \begin{bmatrix} 3 & -3 \\ 0 & 3 \end{bmatrix}$$

$$\circ \circ \quad R \cdot \hat{x} = Q^T b$$

$$\begin{bmatrix} 3 & -3 \\ 0 & 3 \end{bmatrix} \cdot \hat{x} = Q^T \cdot b$$

$$Q = \begin{bmatrix} 1/3 & 2/3 \\ 2/3 & 1/3 \\ -2/3 & 2/3 \end{bmatrix}$$

$$\begin{bmatrix} 1/3 & 2/3 & -2/3 \\ 2/3 & 1/3 & 2/3 \\ -2/3 & 2/3 & 1/3 \end{bmatrix}$$

$$Q^T b = \begin{bmatrix} 1/3 & 2/3 & -2/3 \\ 2/3 & 1/3 & 2/3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix} = \begin{bmatrix} 1/3 + 4/3 - \frac{14}{3} \\ 2/3 + 2/3 + \frac{14}{3} \end{bmatrix} = \begin{bmatrix} -3 \\ 6 \end{bmatrix}$$

$$\circ \circ \quad \begin{bmatrix} 3 & -3 \\ 0 & 3 \end{bmatrix} \hat{x} = \begin{bmatrix} -3 \\ 6 \end{bmatrix}$$

$$\hat{x} = \begin{bmatrix} 3 & -3 \\ 0 & 3 \end{bmatrix}^{-1} \begin{bmatrix} -3 \\ 6 \end{bmatrix}$$

$$\hat{x} = \frac{1}{9} \begin{bmatrix} 3 & 3 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} -3 \\ 6 \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} -9 + 18 \\ 18 \end{bmatrix}$$

$$\circ \circ \quad \boxed{\hat{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}}$$

Lectures 5 to 7 - Introduction to Eigen values & Eigen vectors, Properties of Eigen values and Eigen vectors, Spectral theorem, Symmetric matrices, Cayley-Hamilton theorem statement only.

Let A be a square matrix of order n . If there exists a real or complex number λ and a non-zero vector x such that $Ax = \lambda x$, then x is called the eigen vector of A and λ is its corresponding eigen value.

- The vector x is in null space of $A - \lambda I$.
- The number λ is chosen so that $A - \lambda I$ has a null space.
- $[A - \lambda I]$ must be singular
- $\det\{A - \lambda I\} = 0$ is called the characteristic equation of A and roots of this equation are called characteristic roots or eigen values or latent roots.

→ corresponding to ' n ' distinct eigen values we get ' n ' independent eigen vectors. But when 2 or more eigen vectors are equal, it may or may not be possible to get linearly independent eigen vectors corresponding to repeated roots.

Procedure to find eigen values & eigen vectors :-

- compute the determinant of $A - \lambda I$. With a λ subtracted along the diagonal, this determinant is a polynomial of degree ' n '. It starts with $(-\lambda)^n$.
- Find the roots of this polynomial. The n roots are the eigen values of A .

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→ For each eigen value λ , solve the equation $(A - \lambda I)x = 0$. Since the determinant $|A - \lambda I|$ is zero, there are solutions other than $x = 0$. Those are the eigen vectors.

Example Find eigen values & Eigen vectors :-

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

Solution characteristic equation :- $|A - \lambda I| = 0$

$$|A - \lambda I| = \begin{vmatrix} -\lambda & 1 \\ -2 & -3-\lambda \end{vmatrix} = 0$$

$$(-\lambda)(-3-\lambda) + 2 = 0$$

$$+3\lambda + \lambda^2 + 2 = 0$$

$$\lambda^2 + 2\lambda + \lambda + 2 = 0$$

$$\lambda(\lambda+2) + 1(\lambda+2) = 0$$

$$\lambda = -2 ; \lambda = -1$$

Let's find the eigen vector v_1 associated with the eigen value $\lambda_1 = -1$.

$$A \cdot v_1 = \lambda_1 \cdot v_1$$

$$(A - \lambda_1 I) v_1 = 0$$

$$\begin{bmatrix} -\lambda_1 & 1 \\ -2 & -3-\lambda_1 \end{bmatrix} v_1 = 0$$

$$\boxed{\lambda_1 = -1} \quad \begin{bmatrix} 1 & 1 \\ -2 & -2 \end{bmatrix} \begin{bmatrix} v_{1,1} \\ v_{1,2} \end{bmatrix} = 0$$

$$v_{1,1} + v_{1,2} = 0 \quad -2v_{1,1} - 2v_{1,2} = 0$$

$$-2v_{1,1} = 2v_{1,2}$$

$$v_{1,1} = -v_{1,2}$$

$$\begin{bmatrix} v_{1,1} \\ v_{1,2} \end{bmatrix} = \begin{bmatrix} v_{1,1} \\ -v_{1,1} \end{bmatrix} = K_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

→ Going through the same procedure for the second eigen value :-

$$A \cdot v_2 = \lambda_2 \cdot v_2$$

$$(A - \lambda_2 I) v_2 = 0$$

$$\begin{bmatrix} -\lambda_2 & 1 \\ -2 & -3 - \lambda_2 \end{bmatrix} v_2 = 0$$

$$\text{for } \lambda_2 = -2 \Rightarrow \begin{bmatrix} 2 & 1 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} v_{2,1} \\ v_{2,2} \end{bmatrix} = 0$$

$$2v_{2,1} + v_{2,2} = 0$$

$$-2v_{2,1} - v_{2,2} = 0$$

$$2(-v_{2,1}) = v_{2,2}$$

$$v_{2,2} = -2v_{2,1}$$

$$\therefore \begin{bmatrix} v_{2,1} \\ v_{2,2} \end{bmatrix} = \begin{bmatrix} v_{2,1} \\ -2v_{2,1} \end{bmatrix} = K \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Properties of Eigen Values & Eigen Vectors & Cayley Hamilton theorem :-

Properties of Eigen Values & Eigen Vectors :-

→ If λ is an eigen value of A with x as the corresponding eigen vector then λ^2 is an eigen value of A^2 with the same eigen vector x .

→ For a given Eigen vector x , there corresponds only one eigen value λ .

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- For a given Eigen value, there corresponds infinitely many Eigen vectors.
- $\lambda = 0$ is an eigen value of A , if and only if A is singular i.e. $\det\{A\} = 0$.
- If λ is an eigen value with x as the eigen vector then $\frac{1}{\lambda}$ is an eigen value of A^{-1} provided A^{-1} exists.
- A and A^T have the same eigen values.
- ^{the} Eigen values of a diagonal matrix are just the diagonal elements of the matrix.
- The eigen values of an idempotent matrix are either zero or unity.
- The sum of the eigen values of a matrix is the sum of the elements of the principal diagonal.
- The product of the eigen values of a matrix A is equal to its determinant.

Cayley-Hamilton theorem - Statement

- ~~Ex~~ "Every square matrix satisfies its own characteristic equation".

Example :- Find the eigen values & eigen vectors for the given matrix $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

Solution characteristic equation is $|A - \lambda I| = 0$

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$$\begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix} = 0$$

$$(6-\lambda)[(3-\lambda)^2 - 1] + 2[-2(3-\lambda) + 2] + 2[2 - 2(3-\lambda)] = 0$$

$$(6-\lambda)[\lambda^2 + 9 - 6\lambda - 1] + 2[-6\lambda + 2\lambda + 2] + 2[2 - 6 + 2\lambda] = 0$$

$$(6-\lambda)[\lambda^2 - 6\lambda + 8] + 2[-4\lambda + 2] + 2[-4 + 2\lambda] = 0$$

$$6\lambda^2 - 36\lambda + 48 - \lambda^3 + 6\lambda^2 - 8\lambda - 8\lambda + 4 - 8 + 4\lambda = 0$$

$$-\lambda^3 + 12\lambda^2 - 48\lambda + 44 = 0$$

$$\lambda^3 - 12\lambda^2 + 48\lambda - 44 = 0$$

$$(6-\lambda)[9 + \lambda^2 - 6\lambda - 1] + 2[-6 + 2\lambda + 2] + 2[2 - 6 + 2\lambda] = 0$$

$$(6-\lambda)[\lambda^2 - 6\lambda + 8] + 2[-4 + 2\lambda] + 2[-4 + 2\lambda] = 0$$

$$6\lambda^2 - 36\lambda + 48 - \lambda^3 + 6\lambda^2 - 8\lambda - 8 + 4\lambda - 8 + 4\lambda = 0$$

$$-\lambda^3 + 12\lambda^2 - 36\lambda + 32 = 0$$

$$\lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$$

$$8^3 - 12(64) + 36(8) - 32 = 0$$

$\lambda = 8$ is one of the roots

8	1	-12	36	-32
	0	8	-32	32
	1	-4	4	0

$$\lambda^2 - 4\lambda + 4 = 0$$

$$\lambda^2 - 2\lambda - 2\lambda + 4 = 0$$

$$\lambda(\lambda - 2) - 2(\lambda - 2) = 0$$

$$(\lambda - 2)(\lambda - 2) = 0$$

$$\therefore \lambda = 2, 2$$

∴ The three eigen values are $\lambda_1 = 8, \lambda_2 = 2, \lambda_3 = 2$

when $\lambda_1 = 8$

Consider $(A - \lambda I)x = 0$

$$(6-\lambda)x - 2y + 2z = 0$$

$$-2x + (3-\lambda)y - z = 0$$

$$2x - y + (3-\lambda)z = 0$$

$$\left. \begin{aligned} -2x - 2y + 2z &= 0 \\ -2x - 5y - z &= 0 \\ 2x - y - 5z &= 0 \end{aligned} \right\}$$

$$\textcircled{22} \begin{bmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix} \xrightarrow[R_3 + R_1]{R_2 - R_1} \begin{bmatrix} -2 & -2 & 2 \\ 0 & -3 & -3 \\ 0 & -3 & -3 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} -2 & -2 & 2 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

x, y = pivot variable

z = free variable

$$-3y - 3z = 0 \quad -2x - 2y + 2z = 0$$

$$-3y = 3z$$

$$\boxed{y = -z}$$

$$-2x - 2(-z) + 2z = 0$$

$$-2x + 2z + 2z = 0$$

$$-2x = -4z$$

$$\boxed{x = 2z}$$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2z \\ -z \\ z \end{bmatrix} = z \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$\text{let } z = 1$$

$$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

when $\lambda_2 = 2$

$$\Rightarrow \begin{cases} 4x - 2y + 2z = 0 \\ -2x + y - z = 0 \\ 2x - y + z = 0 \end{cases} \begin{bmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$\begin{bmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} \xrightarrow[2R_3 - R_1]{2R_2 + R_1} \begin{bmatrix} 4 & -2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$4x - 2y + 2z = 0$$

x = pivot variable; y & z = free variables

$$\text{let } y = 1 \text{ \& } z = 0$$

$$4x - 2 = 0$$

$$4x = 2$$

$$x = 1/2$$

$$\text{let } y = 0 \text{ \& } z = 1$$

$$4x + 2 = 0$$

$$4x = -2$$

$$\boxed{x = -1/2}$$

$$4x - 2y + 2z = 0$$

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$$\text{let } z = K_2 \text{ \& } y = K_1$$

$$4x - 2K_1 + 2K_2 = 0$$

$$4x = 2K_1 - 2K_2$$

$$x = K_1/2 - K_2/2$$

$$\therefore x_2 = \begin{pmatrix} K_1/2 - K_2/2 \\ K_1 \\ K_2 \end{pmatrix} = x_3$$

$$\therefore \text{Eigen vector } x = [x_1 \ x_2 \ x_3]$$

where K_1 & K_2 are arbitrary and can take any value.

Example Find the eigen values and eigen vectors for the matrix :-

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ -1 & 1 & -1 \end{bmatrix}$$

Solution :- $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & -1 & 1 \\ 1 & -\lambda & 0 \\ -1 & 1 & -1-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(-\lambda)(-1-\lambda) + 1(1(-1-\lambda)) + 1(1-\lambda) = 0$$

$$(1-\lambda)\{\lambda + \lambda^2\} + 1\{-1-\lambda\} + 1-\lambda = 0$$

$$\cancel{\lambda} + \cancel{\lambda^2} - \cancel{\lambda^2} - \lambda^3 - \cancel{1} - \cancel{\lambda} + \cancel{1} - \lambda = 0$$

$$-\lambda^3 - \lambda = 0$$

$$\Rightarrow \lambda^3 + \lambda = 0$$

$$\lambda(\lambda^2 + 1) = 0 \Rightarrow \text{roots are } \lambda = 0 \text{ \& } \lambda = \pm i$$

(24) Now consider $(A - \lambda I)x = 0$

$$(1 - \lambda)x - y + z = 0$$

$$x - \lambda \cdot y + 0 \cdot z = 0$$

$$-x + y + (-1 - \lambda)z = 0$$

Case i) When $\lambda = 0$ $x - y + z = 0$

$$x = 0$$

$$-x + y - z = 0$$

$$-y + z = 0 \quad y = z$$

$$-y = -z \quad \text{let } z = k_2 \text{ \& } y = k_1$$

$$\boxed{y = z}$$

$$x_1 = \begin{bmatrix} 0 \\ k_1 \\ k_2 \end{bmatrix} ; \text{ let } k_1 = k_2 = 1$$

$$\therefore x_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Case ii) When $\lambda = i$; $(1 - i)x - y + z = 0$

$$-x + y + (-1 - i)z = 0$$

$$x = y(1 - i)$$

$$x - iy = 0$$

$$x = y + (-1 - i)z$$

$$x = iy$$

~~$$y(1 - i) = y + (-1 - i)z$$~~

$$iy = y + (-1 - i)z$$

~~$$y(1 - i) - y = (-1 - i)z$$~~

$$y(i - 1) = (-1 - i)z$$

$$y(i - 1) = (-1 - i)z$$

$$z = \frac{y(i - 1)}{-1 - i} \times \frac{-1 + i}{-1 + i}$$

$$z = \frac{y(-i + i^2 + 1 - i)}{(-1)^2 - (i^2)}$$

$$z = \frac{y(-i - 1 + 1 - i)}{2}$$

$$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} iy \\ y \\ -iy \end{pmatrix} = y \begin{pmatrix} i \\ 1 \\ -1 \end{pmatrix}$$

$$\text{let } y = k_1 = 1$$

$$\therefore x_2 = \begin{bmatrix} i \\ 1 \\ -1 \end{bmatrix}$$

$$\boxed{z = -i \cdot y}$$

Case iii) when $\lambda = -i$

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$$(1+i)x - y + z = 0$$

$$x + iy = 0 \Rightarrow \boxed{x = -iy}$$

$$-x + y + (-1+i)z = 0$$

$$x = y + (-1+i)z$$

$$-iy = y + (-1+i)z$$

$$(-1+i)z = -iy - y$$

$$(-1+i)z = y(-i-1)$$

$$z = y \frac{-i-1}{-1+i} \times \frac{-1-i}{-1-i}$$

$$z = y \frac{\{i + i^2 + 1 + i\}}{(-1)^2 - i^2}$$

$$z = y \frac{\{2i + 1 - 1\}}{2}$$

$$\boxed{z = iy}$$

$$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -iy \\ y \\ iy \end{pmatrix} = y \begin{pmatrix} -i \\ 1 \\ i \end{pmatrix} \quad \text{let } y = k_1 = 1$$

$$\therefore x_3 = \begin{pmatrix} -i \\ 1 \\ i \end{pmatrix}$$

$\therefore$ Eigen vectors corresponding to $\lambda = 0, i, -i$

$$X = [x_1 \ x_2 \ x_3] = \begin{bmatrix} 0 & i & -i \\ 1 & 1 & 1 \\ 1 & -i & i \end{bmatrix}$$

(26)

Example If $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$ Evaluate A^{-1} & A^{-2} using Cayley-Hamilton theorem

Solution:- $|A - \lambda I| = 0$

$$\begin{vmatrix} 4-\lambda & 6 & 6 \\ 1 & 3-\lambda & 2 \\ -1 & -4 & -3-\lambda \end{vmatrix} = 0$$

$$(4-\lambda)\{(3-\lambda)(-3-\lambda)+8\}-6\{1(-3-\lambda)+2\}+6\{-4+(3-\lambda)\}=0$$

$$(4-\lambda)\{-9-3\lambda+3\lambda+\lambda^2+8\}-6\{-3-\lambda+2\}+6\{-4+3-\lambda\}=0$$

$$(4-\lambda)\{\lambda^2-1\}-6\{-\lambda-1\}+6\{-\lambda-1\}=0$$

$$4\lambda^2-4-\lambda^3+\lambda+6\lambda+6-6\lambda-6=0$$

$$-\lambda^3+4\lambda^2+\lambda-4=0$$

$$\boxed{\lambda^3-4\lambda^2-\lambda+4=0}$$

According to CH theorem:- λ to be replaced by A .

$$\therefore A^3-4A^2-A+4I=0 \quad \text{--- (1)}$$

To find A^{-1} , multiply equation (1) by A^{-1}

$$A^{-1}\{A^3-4A^2-A+4I\}=0$$

$$A^2-4A-I+4A^{-1}=0$$

$$4A^{-1}=-A^2+4A+I$$

$$A^{-1} = \frac{-A^2+4A+I}{4}$$

$$A^{-1} = \frac{1}{4} \left[- \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix} \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix} + 4 \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right]$$

$$\therefore A^{-1} = \frac{1}{4} \begin{bmatrix} 1 & 6 & 6 \\ -1 & 6 & 2 \\ 1 & -10 & -6 \end{bmatrix}$$

To find A^{-2} , multiply (1) by A^{-2}
eqn (1)

$$A^3 - 4A^2 - A + 4I = 0$$

$$A^{-2} \{ A^3 - 4A^2 - A + 4I \} = 0$$

$$A - 4I - A^{-1} + 4A^{-2} = 0$$

$$4A^{-2} = -A + A^{-1} + 4I$$

$$A^{-2} = \frac{1}{4} [-A + A^{-1} + 4I]$$

$$A^{-2} = \frac{1}{4} \left[- \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix} + \frac{1}{4} \begin{bmatrix} 1 & 6 & 6 \\ -1 & 6 & 2 \\ 1 & -10 & -6 \end{bmatrix} + \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} \right]$$

We can verify that :-

$$A^{-2} = \frac{1}{16} \begin{bmatrix} 1 & -18 & -18 \\ -5 & 10 & -6 \\ 5 & 6 & 22 \end{bmatrix}$$

(28)

Lectures 8 & 9 - Diagonalization of a matrix, Powers & Products of Matrices

Diagonalization of a Matrix :-

Statement :- If A is a square matrix of order ' n ' has ' n ' linearly independent vectors, then a matrix ' S ' can be found such that $S^{-1}AS$ is a diagonal matrix.

Proof :- Let A be a square matrix of order 3.

Let λ_1, λ_2 and λ_3 be its eigen values and $x_1 = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$
 $x_2 = \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$ and $x_3 = \begin{pmatrix} x_3 \\ y_3 \\ z_3 \end{pmatrix}$ be the corresponding eigen vectors.

Let S be a square matrix $S = [x_1 \ x_2 \ x_3]$
 i.e $S = \begin{pmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{pmatrix}$

multiplying by A , $AS = A[x_1 \ x_2 \ x_3]$
 $= [Ax_1 \ Ax_2 \ Ax_3]$

$$\Rightarrow AS = [\lambda_1 x_1, \lambda_2 x_2, \lambda_3 x_3]$$

$$AS = \begin{bmatrix} \lambda_1 x_1 & \lambda_2 x_2 & \lambda_3 x_3 \\ \lambda_1 y_1 & \lambda_2 y_2 & \lambda_3 y_3 \\ \lambda_1 z_1 & \lambda_2 z_2 & \lambda_3 z_3 \end{bmatrix} = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$

$S \qquad D$

$\therefore AS = SD$

where D is the diagonal matrix $D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$ (29)

$$AS = SD$$

Multiplying both sides by S^{-1}

$$S^{-1}AS = SDS^{-1}$$

$$\boxed{S^{-1}AS = D} \quad \text{or} \quad S^{-1}AS = \Lambda$$

$$\boxed{A = SDS^{-1}}$$

S is invertible because its columns (the eigen vectors) were assumed to be independent.

Note :

1) Any matrix with distinct eigen values can be diagonalizable.

2) Not all matrices has ' n ' linearly independent eigen vectors. Therefore, all the matrices can not be diagonalizable.

Eg $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, here $\lambda_1 = \lambda_2 = 0$. There is only one independent vector. Hence, we cannot construct S .

3) If eigen vectors $x_1, x_2, \dots, x_k$ correspond to distinct eigen values $\lambda_1, \lambda_2, \dots, \lambda_k$, then those eigen vectors are linearly independent.

4) Eigen vector matrix is not unique since if x is an eigen vector corresponding to λ , then kx is also an eigen vector.

(30)

Example Factorize the matrix $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ into $S \Lambda S^{-1}$ and also find $S \Lambda S^{-1}$.

Solution consider $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 1 \\ 1 & 1-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)^2 - 1 = 0 \quad 1 + \lambda^2 - 2\lambda - 1 = 0$$

$$\lambda^2 - 2\lambda = 0$$

$$\lambda(\lambda - 2) = 0$$

$$\lambda = 0 \text{ or } \lambda = 2$$

$$\therefore \Lambda = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$$

To find eigen ~~value~~ vectors :-

$$(A - \lambda I)x = 0$$

$$(1-\lambda)x + y = 0$$

$$x + (1-\lambda)y = 0$$

Case 1 for $\lambda = 0$

$$x + y = 0$$

$$x + y = 0$$

$$x = -y$$

$$\text{let } y = k \text{ or } x = -k$$

$$x_2 = \begin{bmatrix} -k \\ k \end{bmatrix} = k \begin{bmatrix} -1 \\ 1 \end{bmatrix}; \text{ If } k = 1 \Rightarrow x_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Case 2 for $\lambda = 2$

$$-x + y = 0$$

$$x - y = 0$$

$$x = y$$

$$x = y = k; \text{ let } k = 1$$

$$\therefore x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\therefore S = \begin{bmatrix} x_1 & x_2 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}; S^{-1} = \frac{1}{-1-1} \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$$

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$$S^{-1} = \begin{bmatrix} -1/2 & 1/2 \\ 1/2 & +1/2 \end{bmatrix} \quad \Lambda = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$$

$$A = S \Lambda S^{-1} = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -1/2 & 1/2 \\ 1/2 & +1/2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -1/2 & 1/2 \\ 1/2 & +1/2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\therefore \boxed{A = S \Lambda S^{-1}}$$

to find $S \Lambda S^{-1}$

$$\begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} \\ = \begin{bmatrix} 0 & 0 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\therefore \boxed{\Lambda = S \Lambda S^{-1}}$$

Example Find the matrix A whose eigen values are 2, 5 and eigen vectors are $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ & $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

solution $S = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}; S^{-1} = \frac{1}{1} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \quad \Lambda = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$

$$\therefore A = S \Lambda S^{-1} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \\ = \begin{bmatrix} 2 & 5 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 0 & 5 \end{bmatrix}$$

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Example Find the eigen values and Eigen Vectors

of $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ and write two different diagonalizing matrices.

Solution :- $|A - \lambda I| = 0$ $\begin{vmatrix} 1-\lambda & 1 & 1 \\ 1 & 1-\lambda & 1 \\ 1 & 1 & 1-\lambda \end{vmatrix} = 0$

$$(1-\lambda)[(1-\lambda)^2 - 1] - 1[(1-\lambda) - 1] + 1[1 - (1-\lambda)] = 0$$

$$(1-\lambda)[\lambda^2 - 2\lambda + 1] - 1[-\lambda + 1] + 1[\lambda] = 0$$

$$(1-\lambda)[\lambda^2 - 2\lambda] - 1[-\lambda] + 1[\lambda] = 0$$

$$\lambda^2 - 2\lambda - \lambda^3 + 2\lambda^2 + \lambda + \lambda = 0$$

$$-\lambda^3 + 3\lambda^2 = 0$$

$$\lambda^3 - 3\lambda^2 = 0$$

$$\lambda^2(\lambda - 3) = 0$$

$$\boxed{\lambda = 0, 0, 3}$$

Consider $(A - \lambda I)x = 0$

$$(1-\lambda)x + y + z = 0$$

$$x + (1-\lambda)y + z = 0$$

$$x + y + (1-\lambda)z = 0$$

case 1) $\lambda = 0$

$$x + y + z = 0$$

$$x + y + z = 0$$

$$x + y + z = 0$$

$$x = -y - z$$

$$\text{let } y = k_1; z = k_2$$

$$x = -k_1 - k_2$$

$$\text{let } K_1 = K_2 = 1$$

$$\therefore x_1 = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$$

Case 2) $\lambda = 0$

$$\text{let } K_1 = 0 \text{ \& } K_2 = 1 \quad (or) \quad \text{let } K_1 = 1 \text{ \& } K_2 = 0$$

$$\therefore x_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$x_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$$

Case 3) when $\lambda = 3$

$$-2x + y + z = 0$$

$$x - 2y + z = 0$$

$$x + y - 2z = 0$$

$$\begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix} \xrightarrow[2R_3 + R_1]{2R_2 + R_1} \begin{bmatrix} -2 & 1 & 1 \\ 0 & -3 & 3 \\ 0 & 3 & -3 \end{bmatrix} \xrightarrow{R_3 + R_2} \begin{bmatrix} -2 & 1 & 1 \\ 0 & -3 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

x, y = pivot variable & z = free variable

$$-3y + 3z = 0$$

$$-3y = -3z$$

$$\boxed{y = z}$$

$$\text{let } z = K_1$$

$$y = K_1$$

$$-2x + y + z = 0$$

$$-2x + K_1 + K_1 = 0$$

$$-2x = -2K_1$$

$$x = K_1$$

$$\therefore x_3 = \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix}$$

$$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = K_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\text{let } K_1 = 3$$

(34) ∴ Two diagonalizing matrices are ∴

$$S_1 = \begin{bmatrix} -2 & -1 & 3 \\ 1 & 0 & 3 \\ 1 & 1 & 3 \end{bmatrix} \quad \text{or} \quad S_2 = \begin{bmatrix} -2 & -1 & 3 \\ 1 & 1 & 3 \\ 1 & 0 & 3 \end{bmatrix}$$

Powers & Products of Matrices ∴

→ computation of Powers of a Square matrix ∴

→ Diagonalization of a Square matrix A also helps to find the powers of $A, A^2, \dots$ e.t.c

→ We have $D = S^{-1}AS$

$$D^2 = (S^{-1}AS)(S^{-1}AS)$$

~~$$D^2 = S^{-1} \cdot S \cdot A^2 \cdot S$$~~

$$D^2 = S^{-1} \cdot A \cdot \underbrace{S \cdot S^{-1}}_I \cdot A \cdot S$$

$$D^2 = S^{-1} \cdot A^2 \cdot S$$

Premultiplying by S and Postmultiplying by S^{-1}

$$S D^2 S^{-1} = S \cdot S^{-1} \cdot A^2 \cdot S \cdot S^{-1}$$

$$= I \cdot A^2 \cdot I$$

$$= A^2$$

$$\therefore \boxed{A^2 = S \cdot D^2 \cdot S^{-1}}$$

In general, $A^n = S D^n S^{-1}$

where $D^n = \begin{bmatrix} \lambda_1^n & 0 & 0 & 0 & \dots \\ 0 & \lambda_2^n & 0 & 0 & \dots \\ 0 & 0 & \lambda_3^n & 0 & \dots \\ \vdots & & & & \\ 0 & 0 & 0 & \dots & \lambda_n^n \end{bmatrix}$

Example Diagonalize the matrix $A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$ and find one of its square roots. How many square roots will be there?

Solution:- $|A - \lambda I| = 0 \quad \begin{vmatrix} 5-\lambda & 4 \\ 4 & 5-\lambda \end{vmatrix} = 0$

$$\begin{aligned} (5-\lambda)^2 - 16 &= 0 \\ 5^2 + \lambda^2 - 10\lambda - 16 &= 0 \\ \lambda^2 - 10\lambda + 9 &= 0 \\ \lambda^2 - 9\lambda - \lambda + 9 &= 0 \\ \lambda(\lambda-9) - 1(\lambda-9) &= 0 \\ (\lambda-9)(\lambda-1) &= 0 \\ \lambda &= 9 \text{ or } \lambda = 1 \end{aligned}$$

consider $[A - \lambda I]x = 0$

$$\begin{aligned} (5-\lambda)x + 4y &= 0 \\ 4x + (5-\lambda)y &= 0 \end{aligned}$$

case 1) for $\lambda = 1$

$$\begin{aligned} 4x + 4y &= 0 \\ 4x + 4y &= 0 \end{aligned}$$

$$x = -y$$

let $y = k; x = -k$

$$\therefore \begin{pmatrix} x \\ y \end{pmatrix} = k \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

let $k = 1$

$$x_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

case 2 for $\lambda = 9$

$$\begin{aligned} -4x + 4y &= 0 \\ 4x - 4y &= 0 \end{aligned}$$

$$x = y$$

let $y = k = x$

let $k = 1$

$$\therefore x_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$S = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \quad \Delta = 0$$

$$\Delta = \begin{pmatrix} 1 & 0 \\ 0 & 9 \end{pmatrix}$$

$$S^{-1} = \frac{1}{-2} \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

$$S^{-1} = \begin{pmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$$

(36) we know that $A = SDS^{-1}$

(or)

$$A = S \Lambda S^{-1}$$

$$A^{1/2} = S \Lambda^{1/2} S^{-1}$$

$$A^{1/2} = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 3 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

$$A^{1/2} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

we have 4 square roots of $\Lambda^{1/2} = \sqrt{1} = \pm 1$ and also $\sqrt{9} = \pm 3$. we got different values when we consider $\pm 1, \pm 3$.

Example Diagonalize $A = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$ and hence find A^{100} . Show that $A^{100} = A$.

Solution: $|A - \lambda I| = 0 \quad \begin{vmatrix} 1/2 - \lambda & 1/2 \\ 1/2 & 1/2 - \lambda \end{vmatrix} = 0$

$$\left(\frac{1}{2} - \lambda\right)^2 - \frac{1}{4} = 0$$

$$\frac{1}{4} + \lambda^2 - \lambda - \frac{1}{4} = 0$$

$$\lambda^2 - \lambda = 0$$

$$\lambda(\lambda - 1) = 0$$

$$\lambda = 0 \text{ or } \lambda = 1$$

consider $[A - \lambda I]x = 0 \quad \left(\frac{1}{2} - \lambda\right)x + \frac{1}{2}y = 0$

$$\frac{1}{2}x + \left(\frac{1}{2} - \lambda\right)y = 0$$

Case 1) $\lambda = 0$

$$\frac{1}{2}x + \frac{1}{2}y = 0$$

$$\frac{1}{2}x + \frac{1}{2}y = 0$$

$$x = -y$$

$$\text{let } y = K \Rightarrow x = -K$$

$$\therefore \begin{pmatrix} x \\ y \end{pmatrix} = K \begin{pmatrix} -1 \\ 1 \end{pmatrix}; \text{ let } K = 1$$

$$\therefore x_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

Case 2) $\lambda = 1$

$$-\frac{1}{2}x + \frac{1}{2}y = 0$$

$$\frac{1}{2}x - \frac{1}{2}y = 0$$

$$\boxed{x = y}$$

$$\therefore \text{let } y = K; \Rightarrow x = K; \text{ let } K = 1$$

$$\therefore x_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \Rightarrow S = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$$

$$A = S \wedge S^{-1}$$

$$A = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$$

$$A^{100} = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}^{100} \begin{pmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$$

$$A^{100} = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$$

$$A^{100} = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$$

$$\therefore \boxed{A^{100} = A}$$

(38)

Spectral theorem - Statement :-

An $n \times n$ Symmetric matrix A has the following properties :-

- A has ' n ' real eigen values, counting multiplicities.
- The dimension of the eigenspace for each eigen value λ equals the multiplicity of λ as a root of the characteristic equation.
- The eigen spaces are mutually orthogonal, in the sense that the eigen vectors corresponding to different eigen values are orthogonal.
- A is orthogonally diagonalizable.

Spectral Decomposition theorem - Statement :-

Let A be a $n \times n$ matrix. Let $\{u_1, u_2, \dots, u_n\}$ be an orthonormal basis for $\mathbb{R}^n$ consisting of eigen vectors of A with corresponding eigen values $\lambda_1, \lambda_2, \dots, \lambda_n$. Then there exists Symmetric matrices $P_1, P_2, \dots, P_n$ such that the following results hold :-

$$i) A = \lambda_1 P_1 + \lambda_2 P_2 + \dots + \lambda_n P_n$$

$$ii) \text{Rank } P_i = 1 \text{ for all } i$$

$$iii) P_i \cdot P_i = P_i \text{ for all } i \text{ and } P_i \cdot P_j = 0 \text{ for } i \neq j$$

$$iv) P_i \cdot u_i = u_i \text{ for all } i \text{ and } P_i \cdot u_j = 0 \text{ for } i \neq j.$$

Spectral theorem example :-

$$\text{Let } A = \begin{bmatrix} 3 & -4 \\ -4 & -3 \end{bmatrix}$$

TO find eigen values :- $|A - \lambda I| = 0$

$$\begin{vmatrix} 3-\lambda & -4 \\ -4 & -3-\lambda \end{vmatrix} = 0 ; (3-\lambda)(-3-\lambda) - 16 = 0$$

$$-9 + 3\lambda - 3\lambda + \lambda^2 - 16 = 0$$

$$\lambda^2 - 25 = 0$$

$$\lambda^2 = 25 ; \lambda = \pm 5$$

consider $[A - \lambda I]x = 0$

$$(3 - \lambda)x - 4y = 0$$

$$-4x + (-3 - \lambda)y = 0$$

case 1) $\lambda = -5$

$$8x - 4y = 0$$

$$-4x + 2y = 0$$

$$-4x = -2y$$

$$4x = 2y$$

$$2x = y$$

$$x = y/2$$

$$\therefore x_1 = \begin{pmatrix} 1/2 \\ 1 \end{pmatrix}$$

here let $y = 2$

$$x_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

case 2) $\lambda = 5$

$$-2x - 4y = 0$$

$$-4x - 8y = 0$$

$$-4x = 8y$$

$$x = -2y$$

$$\text{let } y = 1$$

$$x = -2$$

$$\therefore x_2 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

The orthonormal eigen vectors are $u_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} -2 \\ 1 \end{pmatrix}$
and $u_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$$P_1 = u_1 \cdot u_1^T = \frac{1}{\sqrt{5}} \begin{pmatrix} -2 \\ 1 \end{pmatrix} \frac{1}{\sqrt{5}} \begin{pmatrix} -2 & 1 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 4 & -2 \\ -2 & 1 \end{pmatrix}$$

$$P_2 = u_2 \cdot u_2^T = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix} \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & 2 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$$

We can verify that $\therefore \boxed{A = \lambda_1 P_1 + \lambda_2 P_2}$